

Theoretical Quantum Optics: Problem Sheet 8

(for 30.06.2025)

23. Phase conjugation (2P)

Consider a single-sided cavity with decay rate γ and a squeezing Hamiltonian

$$H = i\hbar \frac{g}{2} (a^2 - a^{\dagger 2}). \quad (1)$$

This squeezing interaction couples the operators $a(t)$ and $a^\dagger(t)$ and will therefore also mix the out-field $b_{\text{out}}(t)$ with the in-field $b_{\text{in}}^\dagger(t)$.

- (a) Derive the QLEs for the operators $a(t)$ and $a^\dagger(t)$.
- (b) Derive the input-output relations for this system in Fourier space. Show that these relations can be summarized as

$$\begin{pmatrix} b_{\text{out}}(\omega) \\ b_{\text{out}}^\dagger(\omega) \end{pmatrix} = \mathbf{S}(\omega) \begin{pmatrix} b_{\text{in}}(\omega) \\ b_{\text{in}}^\dagger(\omega) \end{pmatrix} \quad (2)$$

and determine the scattering matrix $\mathbf{S}(\omega)$.

- (c) Focus on the resonant case $\omega = 0$ and find a condition under which one obtains the scattering relation

$$b_{\text{out}}(\omega \approx 0) \propto b_{\text{in}}^\dagger(\omega \approx 0). \quad (3)$$

This means that the reflected field is exactly the phase-conjugate of the incoming field.

24. Kerr oscillator (2P)

Consider a driven cavity mode with a Kerr nonlinearity, as modeled by the Hamiltonian (in the rotating frame)

$$H = -\Delta a^\dagger a - \frac{U}{2} a^\dagger a^\dagger a a + i\mathcal{E}(a^\dagger - a). \quad (4)$$

- (a) Derive the QLE for the mode operator $a(t)$, assuming a photon decay rate γ .
- (b) In analogy to the optomechanical system discussed in Sec. 6.1.3, use a classical approximation to derive an equation of motion for the average field amplitude $\alpha(t) = \langle a \rangle(t)$.
- (c) Within the same approximation, discuss (e.g. graphically) the dependence of the steady state photon number $\langle a^\dagger a \rangle_0 \approx |\alpha_0|^2$ as a function of the detuning for different ratios U/κ and $\mathcal{E}/\kappa$. Argue that this system exhibits optical bistability.

25. Kerr oscillator: numerics (2P)

Write a Python program to simulate the steady state of the damped Kerr oscillator from above and evaluate the exact dependence of $\langle a^\dagger a \rangle_0$ on Δ . How does the shape change for fixed $U \approx \gamma$ and increasing driving strength $E \geq \gamma$?

Hint: Start the simulations with $U = 0$ and low driving strength to make sure the results are converged by reproducing the Lorentzian lineshape of the bare cavity.

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Quantum Langevin Equation (QLE):

$$\dot{O}_S = \frac{i}{\hbar} [H_S, O_S] - [O_S, c^\dagger] \left(\frac{\gamma}{2} c + \sqrt{\gamma} b_{\text{in}}(t) \right) - \left(\frac{\gamma}{2} c^\dagger + \sqrt{\gamma} b_{\text{in}}^\dagger(t) \right) [c, O_S]$$

$$\{a^\dagger, a\} = -1$$

$$\begin{aligned} a) \quad \underline{\hat{a}(t)}: \quad \frac{i}{\hbar} [i\hbar \frac{g}{2} (a^2 - \hat{a}^{\dagger 2}), a] &= -\frac{g}{2} [a, a] + \frac{g}{2} [\hat{a}^{\dagger 2}, a] = \frac{g}{2} (\hat{a}^\dagger [\hat{a}^\dagger, a] + [\hat{a}^\dagger, a] \hat{a}^\dagger) \\ &= -\frac{g}{2} 2 \cdot \hat{a}^\dagger = -g \hat{a}^\dagger \end{aligned}$$

$$\begin{aligned} c_{\text{in}} = \hat{a}(t) \Rightarrow [O_S, c^\dagger] \left(\frac{\gamma}{2} c + \sqrt{\gamma} b_{\text{in}}(t) \right) &= [a, a^\dagger] \left(\frac{\gamma}{2} a + \sqrt{\gamma} b_{\text{in}}(t) \right) = \left(\frac{\gamma}{2} \hat{a} + \sqrt{\gamma} b_{\text{in}}(t) \right) \\ &\rightarrow \left(\frac{\gamma}{2} c^\dagger + \sqrt{\gamma} b_{\text{in}}^\dagger(t) \right) [a, a] = 0 \end{aligned}$$

$$\Rightarrow \dot{\hat{a}}(t) = -g \hat{a}^\dagger - \frac{\gamma}{2} \hat{a} - \sqrt{\gamma} b_{\text{in}}(t)$$

$$\begin{aligned} \underline{\hat{a}^\dagger(t)}: \quad \frac{i}{\hbar} [H_S, \hat{a}^\dagger] &= -\frac{g}{2} [\hat{a}^2, \hat{a}^\dagger] + \frac{g}{2} [\hat{a}^{\dagger 2}, \hat{a}^\dagger] = -\frac{g}{2} (\hat{a} [\hat{a}, \hat{a}^\dagger] + [\hat{a}, \hat{a}^\dagger] \hat{a}) \\ &= -g \hat{a} \end{aligned}$$

$$\begin{aligned} c_{\text{in}} = \hat{a}^\dagger(t) \Rightarrow [O_S, c^\dagger] &= 0 \\ &\rightarrow \left(\frac{\gamma}{2} c^\dagger + \sqrt{\gamma} b_{\text{in}}^\dagger(t) \right) [\hat{a}, \hat{a}^\dagger] = \left(\frac{\gamma}{2} \hat{a}^\dagger + \sqrt{\gamma} b_{\text{in}}^\dagger(t) \right) \end{aligned}$$

$$\Rightarrow \dot{\hat{a}^\dagger}(t) = -g \hat{a} - \frac{\gamma}{2} \hat{a}^\dagger - \sqrt{\gamma} b_{\text{in}}^\dagger(t)$$

(a) Derive the QLEs for the operators $a(t)$ and $a^\dagger(t)$.

(b) Derive the input-output relations for this system in Fourier space. Show that these relations can be summarized as

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and determine the scattering matrix $\mathbf{S}(\omega)$.

Fourier

$$\Rightarrow \dot{\hat{a}}(t) = -g \hat{a}^\dagger - \frac{\gamma}{2} \hat{a} - \sqrt{\gamma} b_{\text{in}}(t)$$

$$-i\omega a(\omega) = -(g a^\dagger(\omega) + \frac{\gamma}{2} a(\omega) + \sqrt{\gamma} b_{\text{in}}(t))$$

$$\dot{\hat{a}^\dagger}(t) = -g \hat{a} - \frac{\gamma}{2} \hat{a}^\dagger - \sqrt{\gamma} b_{\text{in}}^\dagger(t)$$

$$-i\omega a^\dagger(\omega) = -(g a(\omega) + \frac{\gamma}{2} a^\dagger(\omega) + \sqrt{\gamma} b_{\text{in}}^\dagger(t))$$

$$b_{\text{out}}(\omega) = b_{\text{in}}(\omega) + \sqrt{\gamma} a(\omega)$$

(all the $b_n(t)$'s here are supposed to be $b_n(\omega)$'s) $b_{out}^+(t) = b_{in}^+(t) + \sqrt{\gamma} a^+(t)$

$$-i\omega a(\omega) = -(g a^+(\omega) + \frac{\gamma}{2} a(\omega) + \sqrt{\gamma} b_{in}(t))$$

$$-i\omega a^+(\omega) = -(g a(\omega) + \frac{\gamma}{2} a^+(\omega) + \sqrt{\gamma} b_{in}^+(t))$$

$$i\omega a(\omega) - \frac{\gamma}{2} a(\omega) = g a^+(\omega) + \sqrt{\gamma} b_{in}(t)$$

$$a(\omega) = \frac{g a^+(\omega) + \sqrt{\gamma} b_{in}(t)}{i\omega - \gamma/2}$$

$$a^+(\omega) = \frac{g a(\omega) + \sqrt{\gamma} b_{in}^+(t)}{i\omega - \gamma/2}$$

$$\rightarrow a(\omega) = \frac{g}{i\omega - \gamma/2} \left(\frac{g a(\omega) + \sqrt{\gamma} b_{in}^+(t)}{i\omega - \gamma/2} \right) + \frac{\sqrt{\gamma}}{i\omega - \gamma/2} b_{in}(t) = \frac{1}{i\omega - \gamma/2} \left(\frac{g^2}{i\omega - \gamma/2} a(\omega) + \frac{g\sqrt{\gamma}}{i\omega - \gamma/2} b_{in}^+(t) + \sqrt{\gamma} b_{in}(t) \right)$$

$$\left((i\omega - \gamma/2) - \frac{g^2}{i\omega - \gamma/2} \right) a(\omega) = \frac{g\sqrt{\gamma}}{i\omega - \gamma/2} b_{in}^+(t) + \sqrt{\gamma} b_{in}(t)$$

$$i\omega - \gamma/2 =: K$$

$$(K - \frac{g^2}{K}) a(\omega) = \frac{g\sqrt{\gamma}}{K} b_{in}^+(\omega) + \sqrt{\gamma} b_{in}(\omega)$$

$$a(\omega) = \frac{1}{K - \frac{g^2}{K}} \left(\frac{g\sqrt{\gamma}}{K} b_{in}^+(\omega) + \sqrt{\gamma} b_{in}(\omega) \right) \rightarrow b_{out}(\omega) = b_{in}(\omega) + \sqrt{\gamma} \left(\frac{g\sqrt{\gamma}}{K^2 - g^2} b_{in}^+(\omega) + \frac{\sqrt{\gamma}}{K - \frac{g^2}{K}} b_{in}(\omega) \right)$$

$$= \frac{K}{K^2 - g^2} \left(\frac{g\sqrt{\gamma}}{K} b_{in}^+(\omega) + \sqrt{\gamma} b_{in}(\omega) \right) = b_{in}(\omega) \left(1 + \frac{\sqrt{\gamma}}{K - \frac{g^2}{K}} \right) + \frac{g\gamma}{K^2 - g^2} b_{in}^+(\omega)$$

$$= \frac{g\sqrt{\gamma}}{K^2 - g^2} b_{in}^+(\omega) + \frac{1}{K - \frac{g^2}{K}} \sqrt{\gamma} b_{in}(\omega)$$

$$\rightarrow a^+(\omega) = \frac{g}{K} \left(\frac{g}{K} a^+(\omega) + \frac{\sqrt{\gamma}}{K} b_{in}(\omega) \right) + \sqrt{\gamma} b_{in}^+(\omega)$$

$$a^+(\omega) \left(K - \frac{g^2}{K} \right) = \frac{\sqrt{\gamma}}{K} g b_{in}(\omega) + \sqrt{\gamma} b_{in}^+(\omega)$$

$$a^+(\omega) = \frac{1}{K - \frac{g^2}{K}} \left(\frac{g\sqrt{\gamma}}{K} b_{in}(\omega) + \sqrt{\gamma} b_{in}^+(\omega) \right)$$

$$= \frac{g\sqrt{\gamma}}{K^2 - g^2} b_{in}(\omega) + \frac{1}{K - \frac{g^2}{K}} \sqrt{\gamma} b_{in}^+(\omega)$$

$$b_{out}^+(t) = b_{in}^+(t) + \sqrt{\gamma} a^+(t)$$

$$\rightarrow \left(\int e^{i\omega t} b_{out}(\omega) d\omega \right)^+ = \left(\int e^{i\omega t} b_{in}(\omega) d\omega \right)^+ + \sqrt{\gamma} \left(\int e^{i\omega t} a(\omega) d\omega \right)^+$$

$$\int e^{i\omega t} b_{out}^+(\omega) d\omega = \int e^{i\omega t} b_{in}^+(\omega) d\omega + \sqrt{\gamma} \int e^{i\omega t} a^+(\omega) d\omega$$

$$\rightarrow b_{out}^+(\omega) = b_{in}^+(\omega) + \sqrt{\gamma} a^+(\omega)$$

$$\rightarrow b_{out}^+(\omega) = b_{in}^+(\omega) + \sqrt{\gamma} \left(\frac{g\sqrt{\gamma}}{K^2 - g^2} b_{in}(\omega) + \frac{1}{K - \frac{g^2}{K}} \sqrt{\gamma} b_{in}^+(\omega) \right)$$

$$= b_{in}^+(\omega) \left(1 + \frac{\sqrt{\gamma}}{K - \frac{g^2}{K}} \right) + \frac{g\gamma}{K^2 - g^2} b_{in}(\omega)$$

$$\Rightarrow \begin{pmatrix} b_{out}(\omega) \\ b_{out}^\dagger(\omega) \end{pmatrix} = \begin{pmatrix} 1 + \frac{\sqrt{\gamma}}{K - \delta^2/K} & \frac{g\gamma}{K^2 - g^2} \\ \frac{g\gamma}{K^2 - g^2} & 1 + \frac{\sqrt{\gamma}}{K - \delta^2/K} \end{pmatrix} \begin{pmatrix} b_{in}(\omega) \\ b_{in}^\dagger(\omega) \end{pmatrix}$$

$\hat{S}$

c)

$$\omega=0 \Rightarrow K = i\omega - \gamma/2 \rightarrow K = -\gamma/2$$

$$\Rightarrow 1 + \frac{\sqrt{\gamma}}{K - \delta^2/K} = 1 + \sqrt{\gamma} \left(\frac{1}{-\gamma/2 + \delta^2/K} \right) = 1 + \sqrt{\gamma} \left(\frac{1}{-\gamma/2 + 2\delta^2/\gamma} \right) = 1 + \sqrt{\gamma} \left(\frac{2}{-\gamma + 4\delta^2/\gamma} \right)$$

$$= 1 + \sqrt{\gamma} \frac{2\gamma}{-\gamma^2 + 4\delta^2} = 1 + \frac{2(\gamma)^{3/2}}{4\delta^2 - \gamma^2}$$

$$\Rightarrow \frac{g\gamma}{K^2 - g^2} = g\gamma \cdot \frac{1}{\frac{\gamma^2}{4} - g^2} = g\gamma \cdot \frac{4}{\gamma^2 - 4g^2} = 4g\gamma \frac{1}{\gamma^2 - 4g^2}$$

we want $b_{out}(\omega \rightarrow 0) \propto b_{in}^\dagger(\omega \rightarrow 0)$

$$b_{out} = S_{11} b_{in} + S_{12} b_{in}^\dagger$$

$$\rightarrow S_{11} \approx 0 \rightarrow 1 \approx \frac{2\gamma^{3/2}}{4\delta^2 - \gamma^2}$$

$$4\delta^2 - \gamma^2 \approx -2\gamma^{3/2}$$

$$\frac{\gamma^2}{2} - 2g^2 \approx \gamma^{3/2}$$

$$\frac{\gamma^2}{2} - 2g^2 \gamma^{-3/2} \approx 0$$

$$\frac{\sqrt{\gamma}}{2} \approx 2g^2 \gamma^{3/2}$$

24. Kerr oscillator (2P)

Consider a driven cavity mode with a Kerr nonlinearity, as modeled by the Hamiltonian (in the rotating frame)

$$H = -\Delta a^\dagger a - \frac{U}{2} a^\dagger a^\dagger a a + i\mathcal{E}(a^\dagger - a). \quad (4)$$

- (a) Derive the QLE for the mode operator $a(t)$, assuming a photon decay rate γ .
 (b) In analogy to the optomechanical system discussed in Sec. 6.3.1, use a classical approximation to derive an equation of motion for the average field amplitude $\alpha(t) = \langle a \rangle(t)$.
 (c) Within the same approximation, discuss (e.g. graphically) the dependence of the steady state photon number $\langle a^\dagger a \rangle_0 \approx |\alpha_0|^2$ as a function of the detuning for different ratios U/κ and $\mathcal{E}/\kappa$. Argue that this system exhibits optical bistability.

$$\{a^\dagger a^\dagger a a, a\} = \{\hat{a}^\dagger \hat{a}^\dagger \hat{a} \hat{a}, \hat{a}\} = \underbrace{\hat{a}^\dagger \hat{a}^\dagger}_{=0} \{\hat{a} \hat{a}, \hat{a}\} + \hat{a}^\dagger \hat{a}^\dagger \{\hat{a} \hat{a}, \hat{a}\} = \hat{a}^\dagger \hat{a}^\dagger (\hat{a} \{\hat{a}, \hat{a}\} + \{\hat{a}, \hat{a}\} \hat{a}) = -2\hat{a}^\dagger \hat{a}^\dagger$$

$$a) \quad \dot{c} = \dot{\hat{a}}(t) \Rightarrow \{0s, c^\dagger\} \left(\frac{\partial}{\partial} c + \gamma b_n(t) \right) = \{a, a^\dagger\} \left(\frac{\partial}{\partial} a + \sqrt{\gamma} b_n(t) \right) = \left(\frac{\partial}{\partial} \hat{a} + \sqrt{\gamma} b_n(t) \right)$$

$$\{a^\dagger, a\} a = -1 \cdot a$$

$$\begin{aligned} \frac{i}{\hbar} [H, a] &= \frac{i}{\hbar} \left(-\Delta [a^\dagger a, a] - \frac{U}{2} [a^\dagger a^\dagger a a, a] + i\mathcal{E} [a^\dagger, a] - i\mathcal{E} [a, a] \right) \\ &= \frac{i\Delta}{\hbar} \hat{a} + \frac{i}{\hbar} U \hat{a}^\dagger \hat{a}^\dagger - \frac{i\mathcal{E}}{\hbar} 1 = \frac{i\Delta}{\hbar} \hat{a} + \frac{i}{\hbar} U \hat{a}^\dagger \hat{a}^\dagger + \frac{\mathcal{E}}{\hbar} \end{aligned}$$

$$\begin{aligned} \Rightarrow \dot{\hat{a}}(t) &= \frac{i\Delta}{\hbar} \hat{a} + \frac{i}{\hbar} U \hat{a}^\dagger \hat{a}^\dagger + \frac{\mathcal{E}}{\hbar} - \gamma \hat{a} - \sqrt{\gamma} b_n(t) \\ &= \underbrace{\hat{a} \left(\frac{i\Delta}{\hbar} - \gamma \right) + \frac{i}{\hbar} U \hat{a}^\dagger \hat{a}^\dagger - \sqrt{\gamma} b_n(t) + \frac{\mathcal{E}}{\hbar}} \end{aligned}$$

$$b) \text{ use } \alpha(t) \approx \langle a \rangle(t) \quad \alpha = \text{reiff}$$

$$\begin{aligned} \Rightarrow \langle \hat{a}^\dagger \hat{a}^\dagger \rangle(t) \approx \alpha^\dagger(t) \alpha^\dagger(t) &= \alpha(t) |\alpha(t)| = r^2(t) e^{i\theta(t)} \\ \Rightarrow \langle b_n(t) \rangle &= 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow \dot{\alpha}(t) &= \alpha(t) \left(\frac{i\Delta}{\hbar} - \gamma \right) + \frac{i}{\hbar} U |\alpha(t)| \alpha(t) + \frac{\mathcal{E}}{\hbar} \\ &= \left(\frac{i\Delta}{\hbar} - \gamma + \frac{i}{\hbar} U |\alpha(t)| \right) \alpha(t) + \frac{\mathcal{E}}{\hbar} \end{aligned}$$

$$\text{here: } \gamma = \kappa$$

$$\text{steady state: } \dot{\alpha}_0 = 0 \Rightarrow \left(\frac{i\Delta}{\hbar} - \kappa + \frac{iU}{\hbar} |\alpha(t)| \right) \alpha(t) e^{i\theta(t)} + \frac{\mathcal{E}}{\hbar} = 0$$

$$\Rightarrow \left(i\Delta - \kappa + iU |\alpha(t)| \right) \alpha(t) = -\mathcal{E}$$

$$(-i\Delta - K - iU|a|^2(t)) a^*(t) = -\varepsilon^*$$

$$\hookrightarrow \overbrace{(i\Delta - K + iU|a|^2(t))}^A \overbrace{(-i\Delta - K - iU|a|^2(t))}^{A^*} |a|^2(t) = |\varepsilon|^2$$

$$|a|^2(t) = \frac{|\varepsilon|^2}{A \cdot A^*} = \frac{|\varepsilon|^2}{(\Delta + U|a|^2)^2 + K^2}$$

$$\Rightarrow |a|(t) = \frac{|\varepsilon|}{\sqrt{A A^*}}$$

$$A \cdot A^* = \Delta^2 - i\Delta K + \Delta U|a|^2 + i\Delta K + K^2 + iK U|a|^2 + U\Delta|a|^2 - iU K|a|^2 + U^2|a|^4$$

$$= \Delta^2 + \Delta U|a|^2 + K^2 + U^2|a|^4 = (\Delta + U|a|^2)^2 + K^2$$

$$\frac{|\varepsilon|^2}{(\Delta + U|a|^2)^2 + K^2} = \frac{\left(\frac{\varepsilon}{K}\right)^2}{\left(\frac{\Delta}{K} + \frac{U|a|^2}{K}\right)^2 + 1}$$

$$U \frac{|a|^2}{K} = \frac{U}{K} \tilde{\varepsilon}^2 - \tilde{n}$$

$$\Rightarrow \text{Redefine } \tilde{\varepsilon} = \frac{\varepsilon}{K}, \quad \tilde{\Delta} = \frac{\Delta}{K}, \quad \tilde{U} = \frac{U}{K}$$

$$\Rightarrow n = \frac{\tilde{\varepsilon}^2}{(\tilde{\Delta} + \tilde{U} n)^2 + 1}$$

Plot: see python